

Formal Specifications (OCL Expressions)

1. “Offerings are unique. In other words, multiple offerings on the same day and time slot must be offered at a different location.”

context Offering

inv: Offering.allInstances () -> forAll (o1, o2 | o1 <> o2 implies (o1.date <> o2.date or o1.timeSlot <> o2.timeSlot or o1.location <> o2.location))

2. “Any client who is underage must necessarily be accompanied by an adult who acts as their guardian.”

context Client

inv: self.age < 18 implies self.guardian->notEmpty() and self.guardian.age >= 18

3. “The city associated with an offering must be one the city’s that the instructor has indicated in their availabilities.”

context Offering

inv: self.city -> includes(self.instructor.availabilities.city)

4. “A client does not have multiple bookings on the same day and time slot.” (for simplicity we consider only identical day and time slots, even though in reality a booking on Monday 3pm – 4pm and another also on Monday 3:30pm – 4:30pm should not be acceptable.)

context Client

inv: self.bookings -> forAll (b1, b2 |
b1 <> b2 implies (b1.date <> b2.date or b1.timeSlot <> b2.timeSlot)

)